1. ARRAYS

Easy

1. Find Largest/Smallest Element in Array
2. Second Largest Element
3. Sum Between i and j
4. Rotate Array by K Elements
5. Non Decreasing Array
6. Equilibrium Index
7. Remove Duplicates from Sorted Array
8. Remove Duplicates from Unsorted Array Using Set
9. Two Sum
10. Pair Sum
11. Check if Array is Palindrome
12. Reverse Array
13. Find Missing Number
14. Count Frequency of Elements
15. Move Zeroes to End
16. Move Negative Numbers to Start
17. Make Unique Array
18. Find Whether Array is Subset of Another Array

Medium

19. Subarray Sum Equals K
20. Count All Subarrays With Given Sum
21. Longest Subarray With Zero Sum
22. Count All Subarrays Having Sum Divisible by K
23. Kadane's Algorithm — Maximum Subarray Sum
24. Merge Intervals

25. Rearrange Positive & Negative Numbers Alternatively
26. Next Permutation
27. Stock Buy and Sell — Single/Multi Transaction
28. Leaders in an Array
29. Trapping Rain Water
30. Product of Array Except Self
31. Dutch National Flag Algorithm
32. Moore's Voting Algorithm
33. N/3 Repeated Number in Array
34. Maximum Sum Circular Array
35. Maximum Subarray Sum After K Concatenation
36. Repeat and Missing Number Array
37. Sum of Two Elements Equals Third
38. Closest Distance Pair
39. 4 Sum Problem
40. Find the Duplicate Number

🔴 **Hard**

41. Maximum Product Subarray
42. Longest Consecutive Sequence
43. Median of Two Sorted Arrays
44. Merge K Sorted Arrays
45. Count Inversions
46. Minimum Swaps to Sort
47. Sliding Window Maximum
48. Maximum Product Count
49. Maximum Sum Rectangle

2. MATRIX / 2D ARRAYS

Easy

1. Matrix Symmetric
2. Spiral Order

Medium

3. Inplace Rotate Matrix 90 Degree
4. Rotate Matrix to Right
5. Set Matrix Zeroes
6. Nth Element of Spiral Matrix
7. Find All Subsquares of Size K
8. Search in a Row-Wise and Column-Wise Sorted Matrix

Hard

9. Matrix Median

3. TWO POINTERS / SLOW & FAST POINTERS

🟢 **Easy**

1. Reverse Array Using Two Pointers
2. Check Palindrome — String/Array

🟡 **Medium**

3. Two Sum — Sorted Array Using 2 Pointers
4. Remove Duplicates from Sorted Array
5. Container With Most Water
6. Intersection of Two Sorted Arrays

🔴 **Hard**

7. 3Sum / 4Sum Problem
8. Trapping Rain Water
9. Linked List Cycle Detection — Floyd's Cycle Algorithm

4. SLIDING WINDOW

🟢 **Easy**

1. Maximum Sum Subarray of Size K
2. Count Occurrence of Anagram

🟡 **Medium**

3. Longest Substring Without Repeating Characters
4. Minimum Size Subarray Sum
5. Longest Repeating Character Replacement
6. Longest Substring with K Distinct Characters

🔴 **Hard**

7. Sliding Window Maximum
8. Minimum Window Substring

5. BINARY SEARCH

Easy

1. Binary Search — Iterative & Recursive
2. Lower Bound / Upper Bound
3. First and Last Occurrence in Sorted Array
4. Square Root Using Binary Search

Medium

5. Search in Rotated Sorted Array
6. Find Peak Element
7. Aggressive Cows
8. Allocate Books
9. Nth Root of Integer
10. Find Minimum in Rotated Sorted Array
11. Koko Eating Bananas

Hard

12. Median of Two Sorted Arrays
13. Painter Partition Problem
14. Find Kth Element in Two Sorted Arrays
15. Find Kth Element

6. SORTING

Easy

1. Bubble Sort
2. Insertion Sort
3. Selection Sort

Medium

4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Insertion Sort on Linked List
8. Quick Sort on Linked List

Hard

9. Counting Sort
10. Bucket Sort
11. Radix Sort
12. External Merge Sort

7. STRINGS

Easy

1. Check Palindrome String
2. Remove Vowels / Spaces
3. Find ASCII Value of a Character
4. Capitalize First & Last Character of Each Word
5. Find Frequency of Characters
6. Reverse Words and Characters
7. Splitting String into Words
8. Reverse String Word Wise
9. First Non-Repeating Character in String
10. Check Permutation

Medium

11. Longest Common Prefix
12. Longest Substring Without Repeating Characters
13. Valid Anagram
14. Group Anagrams
15. Reverse Words in a Sentence
16. Longest Palindromic Substring
17. Count and Say Problem
18. String Encoding
19. Beautiful Strings
20. Minimum Parenthesis
21. Next Smallest Palindrome
22. Find Kth Character of Decrypted String
23. Multiply Strings
24. Check Subsequence

25. Sum of Zeroes
26. String to Integer (atoi)

Hard

27. Minimum Window Substring
28. Regular Expression Matching
29. Word Break
30. Edit Distance — Levenshtein Distance
31. Rabin-Karp / KMP String Matching
32. Minimum Operations to Make String Equal

8. STACK & QUEUE

🟢 Easy

1. Implement Stack Using Array
2. Implement Stack Using Linked List
3. Implement Queue Using Array/Linked List
4. Reverse Queue
5. Valid Parentheses

🟡 Medium

6. Implement Queue Using 2 Stacks
7. Implement Stack Using 2 Queues
8. Min Stack
9. Next Greater Element
10. Stock Span Problem
11. Sort a Stack

🔴 Hard

12. Largest Rectangle in Histogram
13. Online Stock Span

9. LINKED LIST

Easy

1. Reverse a Linked List
2. Mid Point in Linked List
3. Add Two Linked Lists
4. Check Linked List is Palindrome
5. Merge Two Sorted Linked Lists

Medium

6. Insertion Sort on Linked List
7. Delete Kth Node From End
8. Detect and Remove Cycle
9. Swap Nodes in Pairs
10. Append Nodes
11. Segregate Odd Even
12. K Reverse Linked List
13. Rearrange Linked List
14. Clone a Linked List With Random and Next Pointer

Hard

15. Quick Sort on Linked List
16. Sorted Linked List to Balanced BST

10. TREES

Easy

1. Preorder / Inorder / Postorder Traversal
2. Level Order Traversal — BFS
3. Inorder Traversal Using Stacks

Medium

4. Height of Tree
5. Diameter of Binary Tree
6. Lowest Common Ancestor
7. Mirror Tree
8. Check Balanced Tree
9. ZigZag Order Traversal
10. Left View of Binary Tree
11. Top View of Binary Tree
12. Bottom Right View of Binary Tree
13. LCA of 3 Nodes
14. Time to Burn Tree
15. Merge Two Binary Trees
16. Flatten Binary Tree to Linked List

Hard

17. Serialize and Deserialize Binary Tree
18. Construct Binary Tree From Inorder & Preorder
19. Boundary Traversal of Binary Tree
20. Binary Tree to Doubly Linked List
21. Size of the Largest BST

11. BINARY SEARCH TREE (BST)

Easy

1. Check if Binary Tree is BST

Medium

2. LCA of Two Nodes in BST
3. BST Delete
4. Kth Smallest Element in BST
5. Predecessor and Successor in BST
6. Pair Sum in BST
7. Remove Keys Outside Given Range
8. BST Iterator
9. Merge Two BSTs
10. Fix BST